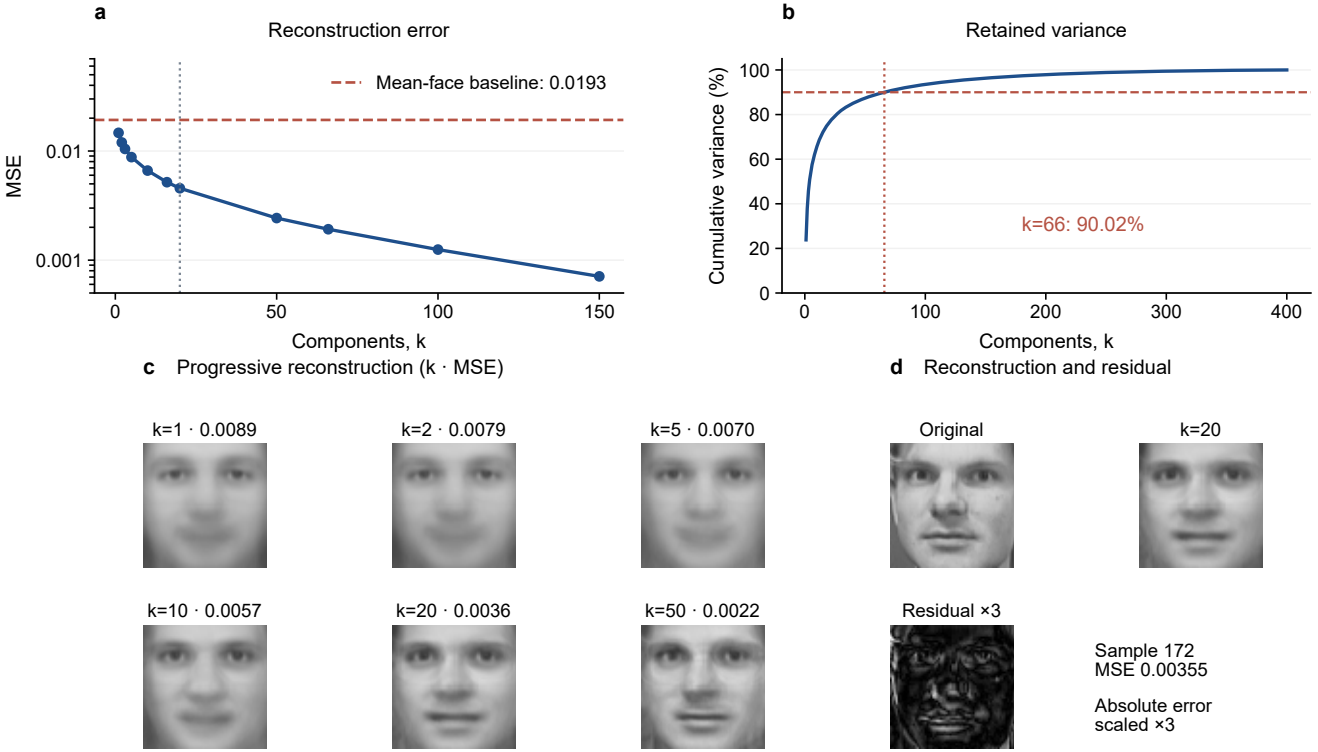

a
Reconstruction error
Mean-face baseline: 0.0193
MSE
0.01
0.001
0
50
100
150
Components, k
b
Retained variance
Cumulative variance (%)
100
80
60
40
20
0
k=66: 90.02%
0
100
200
300
400
Components, k
c Progressive reconstruction (k · MSE)
k=1 · 0.0089
k=2 · 0.0079
k=5 · 0.0070
k=10 · 0.0057
k=20 · 0.0036
k=50 · 0.0022
d Reconstruction and residual
Original
k=20
Residual ×3
Sample 172
MSE 0.00355
Absolute error
scaled ×3